

Evidence of the pair instability gap from black hole masses

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 1:

Reviewer comments:

Referee #1

(Remarks to the Author)

Summary

The manuscript “Evidence of the pair-instability gap in the distribution of black hole masses” by H. Tong and co-authors presents an analysis of the black hole (BH) mass distribution inferred from LVK detections, claiming robust evidence for the pair-instability (PISN) mass gap, with its lower edge around $45 M_{\odot}$. By separating the component masses of binary black hole mergers, the authors find that the secondary-mass (less massive BH) distribution terminates between ~ 40 – $50 M_{\odot}$ and reappears above $\sim 100 M_{\odot}$. They further report a relation between BH mass and effective spin (χ_{eff}), which they interpret as evidence that BHs within the gap are second-generation merger remnants. The topic is timely and of broad interest, and the analysis is important for interpreting gravitational-wave data. The manuscript is clearly written and addresses a significant question. However, in my view, the evidence presented does not justify the strong claim of a robust detection of the pair-instability gap. Several aspects of the analysis appear model-dependent, the GW data seem to contradict key statements (e.g., χ_{eff} distribution), and the discussion of observational biases, events with two components in PSN mass gap and alternative explanations is incomplete.

Major Concerns

1. Distribution of highly spinning BHs does not appear isotropic

The LVK population analysis (<https://arxiv.org/pdf/2508.18083>), for example in their Fig. 13, shows that events within the PISN mass gap preferentially have positive effective spins. This appears to contradict the authors’ claim that the population of highly spinning BHs within the mass gap is isotropic, with an effective spin distribution symmetric around zero. Instead, there is a clear preference for positive effective spin values. This observation is inconsistent with a purely dynamical formation scenario such as formation in globular clusters (as suggested by the authors) where one would expect a symmetric distribution of effective spins, including events with high negative spins. In fact, no confidently negatively spinning events are observed among the current LVK detections.

In fact Fig. 3 of this work presents only absolute value of effective spin, which obscures any sign asymmetry in the effective-spin distribution. I recommend the authors replace or supplement the current panel with the signed χ_{eff} distribution and discuss any observed asymmetry and its implications.

2. Events within and near the putative gap

Some reported LVK events include both BH components within the nominal PISN mass gap, for example GW231028_153006 and GW231005_021030. The manuscript does not discuss how these systems fit within the proposed framework, nor whether they challenge the existence or sharpness of the mass gap.

3. Absence of a feature at $45 M_{\odot}$ in the primary distribution

The primary-mass distribution shows a feature around $35 M_{\odot}$, not at $45 M_{\odot}$, according to both recent and previous LVK analyses. It is unclear why some PISN-driven feature would not appear consistently in both components around $45 M_{\odot}$, as one might expect.

4. Potential selection effects and χ_{eff} bias

The authors should consider and explain whether the apparent trend of χ_{eff} spanning a wider range at higher masses could arise from observational biases. For example, high-mass binaries have shorter inspiral durations in the LVK band, which naturally leads to broader posteriors on χ_{eff} estimates.

5. Posteriors and high-spin claims

If the events within the mass gap originate from second-generation mergers, the individual BH spins are expected to be centered around $a \approx 0.7$ (Gerosa & Berti 2017). The authors do not discuss the posteriors of the individual spins for these PISN events. I had a look and some cases shows peaks at lower values, around 0.4, making them unlikely products of second-generation mergers.

6. Model testing strategy

The models explored in the manuscript appears tuned to reproduce the authors' preferred scenario (e.g., by effectively forcing a mass gap). To robustly assess the strength of the evidence, the authors should test a broader suite of alternative models. Crucially, the authors should quantify the relative support for each hypothesis and e.g., present the results in a summary table. Without such systematic model comparison, it is difficult to determine whether the data truly favor the proposed interpretation or whether the preferred model merely reflects prior/model choices.

Minor Comments

- The upper-mass constraint (Extended Fig. 1) relies just on a single event.
- Alternative formation pathways. The manuscript does not discuss well other formation channels, such as binary or triple interactions (possible spin-up, impact on secondary distribution). The authors do not explain well why the alternative scenarios are disfavored compared with the second-generation merger interpretation.

Recommendation

The manuscript addresses an important question and presents an interesting interpretation of LVK data. Once major issues are addressed, the study would be well suited for publication in other peer-reviewed journal. Nevertheless, in my opinion, the current analysis does not provide sufficiently robust evidence to support the claim of a distinct PISN mass gap and proposed scenario. I therefore do not recommend publication in Nature. However, I suggest that the editor consider giving the authors a chance to address all the major concerns, if the opinion of the other reviewer is positive.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

This manuscript describes the results of a hierarchical Bayesian inference analysis of the binary black hole (BBH) merger events from the fourth LVK catalog (GWTC-4) that identifies strong evidence for a completely empty gap in the secondary mass distribution that the authors interpret as the signature of pair-instability supernovae. While previous analyses have looked for such a gap in the primary mass distribution, this is the first population analysis that searches for this signature in the secondary mass distribution. A total dearth of BBH events with masses $\sim > 45 M_{\text{sun}}$ is a clean imprint of the predicted pair-instability supernova process, by which $\sim 100\text{--}250 M_{\text{sun}}$ stars are completely disrupted, leaving behind no remnant black holes in this mass range. The fact that this feature is only present in the secondary mass distribution is interpreted by the authors as evidence for hierarchical mergers polluting the corresponding gap in the primary mass distribution. The manuscript is generally well-written, and these results are timely and significant, with wide-reaching implications. The authors highlight the constraints that can be placed on the contribution of hierarchical mergers to the underlying BBH population and on the ^{12}C ^{16}O nuclear reaction rate and consider several alternative population models to verify the robustness of their results.

I have one significant concern that the authors should address before I recommend the manuscript for publication in Nature. The fact that the mass model used requires there to be a gap in the secondary mass distribution is hidden in footnote 3 of the Methods section. While I understand that the Bayes factor gives a strong preference for the model with the gap, Bayes factors are hard to interpret since they are prior- and parameterization-dependent. The analysis should be repeated with a prior that allows for the width of the gap to go down to 0 M_{sun} for a robust statement to be made on whether or not the data prefer the presence of a gap. An even better choice would be a slab-spike prior that gives equal prior support to the no-gap and gap hypotheses. I understand that this is partially addressed by the test with the notch filter of variable depth in Section 1.3, but the prior probability for an exactly continuous distribution is still 0 in this case. A plot of the posterior on the gap width, w_g , should also be included, as it's hard to tell how strongly a nonzero gap width is preferred from a plot of the joint posterior on the upper and lower gap edges. Such a prior choice that allows for a gap rather than enforcing its presence would make a much stronger argument for the gap model being preferred by the data. The Bayes factors alone are not convincing when the model enforces the presence of the gap.

Besides this main concern, I have a few detailed comments that I would ask the authors to consider.

- The discussion of GW190521 suggests that the most likely interpretation of this event given the inferred population model is as a straddling binary. The authors also find that their constraint on the upper edge of the mass gap hinges on GW231123. However, looking at Fig 3, it seems like there are two events with m_2 above the upper gap edge. Is this second event GW190521? Is GW190521 still a straddling binary when GW231123 is omitted from the population?
- It should be clarified in the main text that the population model fits m_1/q rather than m_1/m_2 . This was not clear until the Methods section, where Eq 3 is presented. I also wonder what the resulting mass ratio distribution is, if there is such a strong gap in m_2 . Is it inconsistent with the typical power-law once marginalized over m_1/m_2 ? It would be interesting to see the PPD on mass ratio inferred under this population model.
- Is there a typo in the x axis label of Extended Fig. 6, which just reads "The 99"? In any case, this quantity is missing units of M_{sun} .
- The manuscript would benefit from a brief explanation of the astrophysical S-factor on which constraints are placed. Since many readers will not be nuclear physicists, it would be helpful to know what this parameter controls and how it affects the ^{12}C ^{16}O nuclear reaction rate.

(Remarks on code availability)

Referee #3

(Remarks to the Author)

In this paper, the authors explore the emergence of the black hole mass gap in the recently released LIGO data. It has long been thought that stellar-mass black holes between $\sim 45 M_{\odot}$ - $135 M_{\odot}$ cannot form from the collapse of single stars, as pair-production in the cores of these massive stars would produce large pulsations, limiting the final mass of the remnant. However, this feature has been noticeably absent from the primary mass distribution observed by LIGO to date. By assuming that the primary mass distribution is polluted by hierarchical mergers in star clusters, the authors instead fit the gap to the secondary masses, which in most theoretical works are the lower-mass BHs formed from stellar collapse. In doing so, the authors both identify the gap as starting at $45 (+5/-4) M_{\odot}$, and use that result to measure the reaction rate $S(C12\text{-}\alpha)$ at $256 (+197/-104) \text{ keV b}$.

I find the manuscript to be well written, accessible to someone outside the field, and a clear indication of the power of LIGO to measure fundamental physics using gravitational waves. Figure 1, in particular, is a very clear and very compelling way of showing how the mass gap arises naturally out of the current LIGO data and a simple picture of hierarchical mergers. I have a few points related to the astrophysics of hierarchical mergers, as well as the PISN physics, that I would ask the authors to address before publication.

Notes on Hierarchical Mergers:

The analysis in this paper depends very strongly on the assumption that $1G+2G$ BBHs are fairly common, while $2G+2G$ mergers are relatively rare. But most formation scenarios do predict *some* population of $2G+2G$ mergers, suggesting that, with more detections, the gap in m_2 will eventually be filled in as well. If one were to assume these mass-gap BHs originated from globular clusters, as is suggested here, then it is only a matter of luck that a $2G+2G$ merger has not been observed to pollute the gap. I would ask the authors to expand more on when (in terms of number of detections) we expect the m_2 gap to be similarly polluted, and whether this is a function of the specific formation channels (e.g. globular clusters, AGN disks, etc.).

Notes on PISN physics:

The authors provide a brief but good theoretical explanation of how the PISN process works and relating it to the features in the black hole mass spectrum. This is a satisfactory introduction to the theory of the problem.

The authors' model constrains the lower edge of the PISN mass gap for secondary (m_2) objects to $45 (+7/-6) M_{\odot}$. They remark that this is "remarkably consistent" with previous theoretical predictions. It should be noted that this prediction is highly reaction rate-dependent. Later figures in the paper reference the deBoer 2017 estimate for $12C$ alpha, but it should be noted that these theoretical papers use older rate estimates (Buchmann 1996; <https://ui.adsabs.harvard.edu/abs/1996ApJ...468L.127B/abstract>, Kunz et al., 2002; <https://ui.adsabs.harvard.edu/abs/2002ApJ...567..643K/abstract>) which disagree significantly. I contend that Figs. 4 and 8 should plot the S_{300} for these rates as well as the deBoer rate, if these theoretical predictions are going to be submitted for comparison.

Furthermore, the authors' constraints for the S -factor of $12C$ alpha diverges from the deBoer rate estimate significantly ($>+3\sigma$). Previous computational work on this problem using the median deBoer rate found a lower edge for the BH mass gap of $\sim 60 M_{\odot}$ and did not compute models using reaction rates as high as the authors' estimate (Frag 2023). This fact itself warrants discussion, and it would be enlightening to see how the authors' estimates overlap with other sources for the $12C$ alpha rate.

Minor/Textual comments:

Overall, the paper relies a little too heavily on citations to other analytic models. It would improve readability for a broader audience if the main features of these models were stated clearly (i.e. if some of the description from section 1.2 were moved into the main text).

(Remarks on code availability)

Referee #4

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports.

(Remarks on code availability)

Version 2:

Reviewer comments:

Referee #1

(Remarks to the Author)

The authors have carried out a solid and careful analysis, exploring the robustness of a putative mass gap in the secondary-mass distribution of binary black holes. In my view, they have done essentially what is possible with the current gravitational-wave catalog. The manuscript presents a coherent and plausible scenario that, given the present dataset, offers a promising interpretation of the observations. The additional tests included in the revised version provide meaningful cross-checks and further support the conclusion that a genuine trough or gap in m_2 is preferred by current detections.

That said, several aspects of the interpretation remain limited more by the quality and quantity of the available data than by any shortcoming in the authors' analysis. For example, the evidence for a fully isotropic spin distribution in the high-mass population is consistent with isotropy but does not demonstrably favour it over mildly positive-shifted alternatives; the individual spin posteriors for putative second-generation black holes are too broad to support the characteristic $a \sim 0.7$; and the absence of a feature at $m_1 \approx 45 M_\odot$ can be accommodated within their scenario, but such an explanation is intrinsically model-dependent.

While the pair-instability interpretation proposed by the authors is certainly a promising explanation, I am not fully convinced that, given the still small-number statistics, alternative scenarios can be ruled out. In particular, a standard initial mass function (IMF, which limits the number of massive stars), either alone or combined with non-conservative binary interactions, could produce the reported cut-off in the secondary-mass distribution, while high-mass χ_{eff} features could also be shaped by selection effects or other astrophysical mechanisms. Similar caveats apply to the flexibility in assigning several candidate events to the putative PSN mass gap.

Moreover, I note that a recent preprint (Ray & Kalogera 2025, arXiv:2510.18867) appears to challenge some of the claims regarding a sharp mass gap in the secondary-mass distribution. The authors could strengthen their manuscript by explicitly discussing this work, clarifying potential points of agreement or tension, and explaining whether their conclusions remain robust in light of these results.

I therefore recommend that the manuscript be revised to:

1. Moderate the strength of claims regarding the definitiveness of the PSN interpretation or try to quantify alternative explanations (e.g., the role of IMF in m_2 limit);
2. Include a discussion of the recent preprint by Ray & Kalogera (2025, arXiv:2510.18867), addressing any potential tension or reconciliation with their results.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

I thank the authors for addressing my previous comments and concerns. I find the new posterior on the gap width shown in Extended Fig. 1 to be particularly compelling evidence for the presence of the mass gap in the m_2 distribution. The current manuscript is clear, provides strong justification and validation of the mass gap conclusion, and presents one of the most significant results in gravitational-wave population inference to date, revealing a feature that has long been predicted in astrophysical models. I recommend the manuscript for publication in Nature. Below, I include some stylistic comments for the authors' consideration.

- The revised introduction of the parameterization used for the population model is confusing. The main population model considered in this work does not directly model the distribution of $\pi(m_1, m_2)$, so keeping this in L108 seems misleading to me. Waiting to define the mass ratio q until the sentence after it is introduced also comes across as disjointed. I encourage the authors to revise these sentences to improve the clarity and flow.
- I would encourage the authors to add the fact that the absence of a gap is ruled out at high significance (by eyeballing the new Extended Figure 1, it seems like $> 5\sigma$) to the paragraph at the bottom of page 3. This, to me, is much more convincing than a $\ln B$ preference of ~ 4.5 .
- Some typos in the new L187: While hierarchical mergers in dense stellar clusters are usually predicted to have isotropic spin "orientations" due to dynamical assembly, preferentially aligned spins can arise from black holes "in AGN disks" (?) or interactions between black holes and stars in clusters [49, 83].
- L202 quotes $\ln B > 8$ as the threshold for "strong support," but the Bayes factors between population models in the text and in the new Table 1 are not given as $\ln B$ s. Since the log is easier to interpret in terms of significance, I recommend that the authors switch to using $\ln B$ s to compare the preference between population models.
- Some typos in L300: "The slopes of the two bands of *the* broken power law distribution are consistent with each other. We interpret the initial requirement of a second power law in *the* m_1 distribution in current data as a projection of the dearth of high m_2 events. *Instead,* ..."
- Please add $\ln B$ s comparing the evidence for the pairing mass model and spin transition models (Eq 7/8) against the other models to Table I.
- I suggest moving the new Section 1.9 to immediately follow Section 1.6, since they are both focused on the spin transition model and the spin of the high-mass subpopulation.

(Remarks on code availability)

Referee #3

(Remarks to the Author)

The authors have addressed my main questions, and I believe the manuscript is now acceptable.

(Remarks on code availability)

Referee #4

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports.

(Remarks on code availability)

Version 3:

Reviewer comments:

Referee #1

(Remarks to the Author)

The authors have addressed my concerns by providing additional discussion of the limitations. While alternative scenarios remain and I believe the current data are not yet sufficient to fully confirm the pair instability gap interpretation, the manuscript presents a robust analysis and a compelling interpretation of the gravitational wave detection data.

It is interesting and has the potential for broad impact in the field and related areas. Therefore, I believe I can now recommend it for publication.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

I thank the authors for considering my previous suggestions and am happy to recommend the manuscript for publication in Nature in its current form.

(Remarks on code availability)

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We thank the referees for their thoughtful comments. We reply below with the referee comments in black and our own replies in purple.

Referee #2 (Remarks to the Author):

Major Concerns

1. Distribution of highly spinning BHs does not appear isotropic

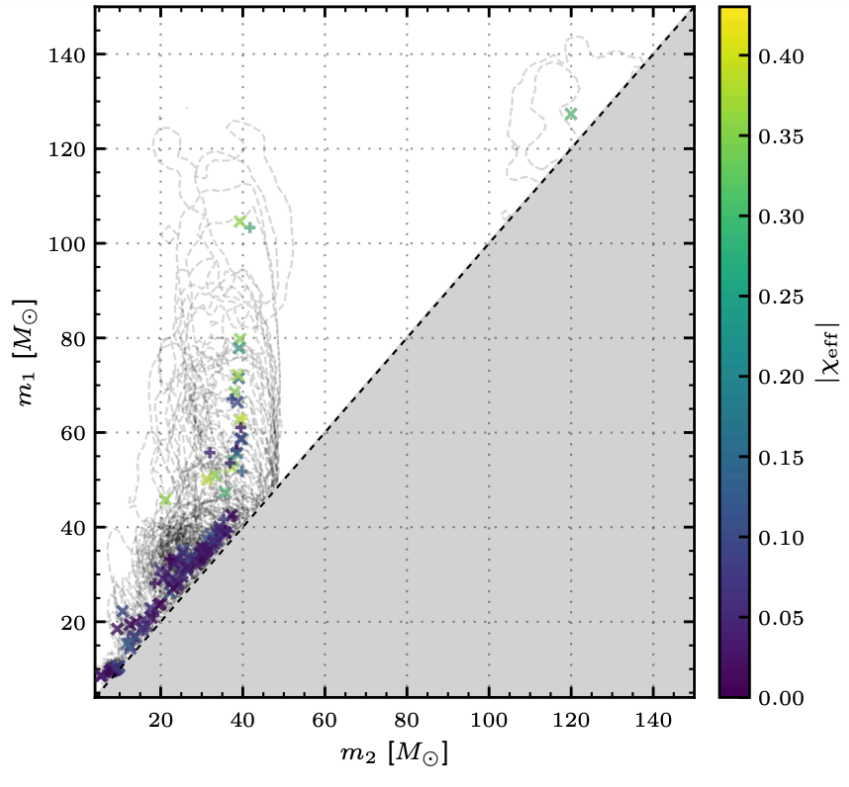
The LVK population analysis (<https://arxiv.org/pdf/2508.18083>), for example in their Fig. 13, shows that events within the PISN mass gap preferentially have positive effective spins. This appears to contradict the authors' claim that the population of highly spinning BHs within the mass gap is isotropic, with an effective spin distribution symmetric around zero. Instead, there is a clear preference for positive effective spin values. This observation is inconsistent with a purely dynamical formation scenario such as formation in globular clusters (as suggested by the authors) where one would expect a symmetric distribution of effective spins, including events with high negative spins. In fact, no confidently negatively spinning events are observed among the current LVK detections.

In fact Fig. 3 of this work presents only absolute value of effective spin, which obscures any sign asymmetry in the effective-spin distribution. I recommend the authors replace or supplement the current panel with the signed χ_{eff} distribution and discuss any observed asymmetry and its implications.

Figure 13 of the LVK population analysis (arxiv/2508.18083) shows the distribution of the effective inspiral spin parameter χ_{eff} for binaries with at least one mass in the 30-40 M_{sun} range, but our analysis focuses on binaries where the primary mass is $>45 M_{\text{sun}}$. Thus, comparing Figure 13 to our results is a bit of an apples-to-oranges comparison. Antonini et al (arxiv/2506.09154) looked into the symmetry of the effective spin distribution of the high mass mergers, and found that the χ_{eff} distribution for the high-mass BBH population is consistent with being symmetric about zero; see their Section III.C.

It's difficult to measure individual events with clearly negative χ_{eff} . The signature is relatively weak, and a degeneracy between (q, χ_{eff}) creates a long tail consistent with $\chi_{\text{eff}} > 0$. (The tails are asymmetric because q is bounded at $q=1$.) Additionally, there's a selection effect that makes it slightly easier to detect events with $\chi_{\text{eff}} > 0$ (see <https://arxiv.org/abs/1805.03046>). Thus, it's difficult to eyeball individual-event posteriors for χ_{eff} and draw conclusions. A better option is to use hierarchical Bayesian inference, which takes into account both the (q, χ_{eff}) degeneracy and selection effects. Thus, we stand by the spin statements in our paper. That said, we note that there are individual events that exhibit signatures of $\chi_{\text{eff}} < 0$ including GW191109, and the just-published low-mass event GW241110.

Here we show a version of the Figure 3 in which the events with negative median χ_{eff} posterior are highlighted by "+" marker and positive χ_{eff} events are represented by "x" marker.



To further address this concern, we've rerun our analysis using a more flexible spin model following Antonini et al, which allows the χ_{eff} distribution for high mass mergers to have free upper and lower boundaries. This allows us to test whether the distribution is symmetric around zero or not. Similarly to Antonini et al, we find that the spin distribution is consistent with an isotropic distribution. We've added subsection 1.9 to discuss this result.

However, even if we did observe an asymmetry, that could well be consistent with our interpretation of a gap contaminated with hierarchical mergers. Hierarchical mergers with aligned spins can also arise from AGN; see, e.g., arxiv/2309.15213 by McKernan & Ford who have written extensively on this topic. Moreover, exciting work by Kiroğlu et al suggests that stellar collisions may even lead to asymmetry in the χ_{eff} distribution for mergers in globular clusters; see arxiv/2410.01879.

We've expanded our discussion on L186 in order to make this point more clear.

While hierarchical mergers in dense stellar clusters are usually predicted to have isotropic spin orientation due to dynamical assembly, preferentially aligned spins can arise from black holes AGN or interactions between black holes and stars in clusters [49, 83]. In Section 1.9, we test the symmetry of the spin distribution for the high mass mergers and show it is consistent with being symmetric around zero, matching the prediction of 2G+1G mergers from dense stellar clusters well.

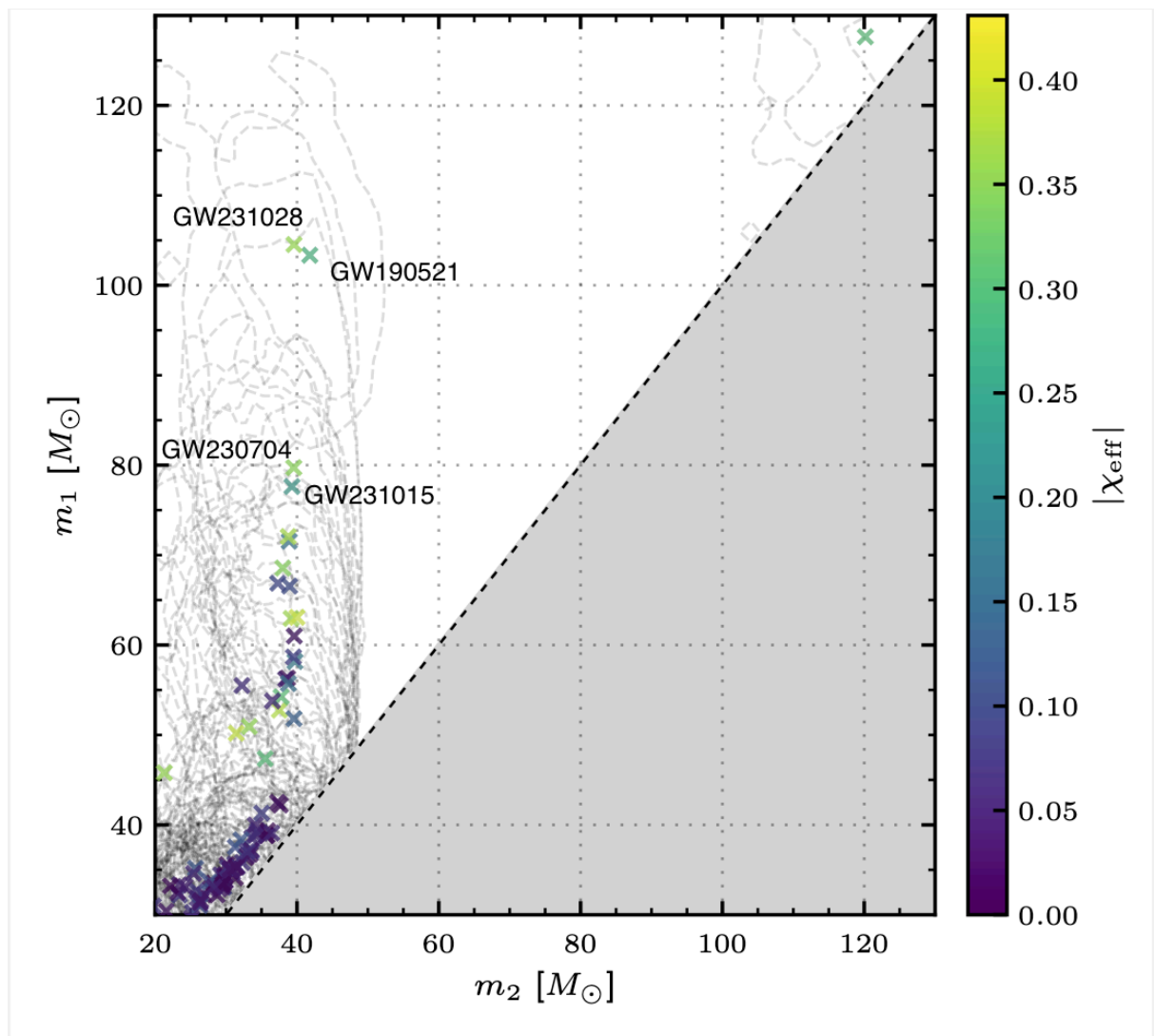
2. Events within and near the putative gap

Some reported LVK events include both BH components within the nominal PISN mass gap, for example GW231028_153006 and GW231005_021030. The manuscript does not discuss

how these systems fit within the proposed framework, nor whether they challenge the existence or sharpness of the mass gap.

We've made an annotated version of our Fig. 3 below to highlight GW231028 and GW231015. Both events are consistent with the gap. From the default parameter estimation, the event GW231028 is consistent with $m_2 < 45$ at 46% credibility while GW231015 is consistent with $m_2 < 45$ at 35% credibility. Of course, these posteriors are obtained using the fiducial LVK priors, which are uniform in the detector-frame (m_1, m_2) plane. The astrophysical distribution of black hole mass declines near 45 Msun. Thus, any reasonable astrophysical prior would serve to make these two events even more consistent with the gap.

Moreover, in Section 1.4, we show a purely data-driven result without any mass model. The purple dots in the “ π stroke” plot in Extended Fig. 4 represent the best-fit mass distribution—the one that maximises the population likelihood. The π stroke fit is consistent with our mass gap model result. This shows unequivocally that the gap we find does not arise entirely from model misspecification; the data support a distribution with a gap between $m_2 = (45, 130)$ Msun.



3. Absence of a feature at $45 M_{\odot}$ in the primary distribution

The primary-mass distribution shows a feature around $35 M_{\odot}$, not at $45 M_{\odot}$, according to both recent and previous LVK analyses. It is unclear why some PISN-driven feature would not appear consistently in both components around $45 M_{\odot}$, as one might expect.

Indeed, one of the predictions of the theory of pair instability supernovae is that there should be a gap in the distribution of black hole masses. However, with the LVK continually detecting massive events in the pair instability gap, researchers have suggested for some time that the gap may be contaminated by hierarchical mergers. (This discussion goes back to at least GW190521, which had a total mass of $150 M_{\text{sun}}$.) Thus, it's not surprising that there's not an obvious feature around $m_1 = 45 M_{\text{sun}}$ if the second-generation black holes have a high rise around the same mass scale to compensate for the rate drop of first-generation black holes; see Fig. 1 of arxiv/2509.04637 for an example of such a model.

Interestingly, the blue curve in Fig. 3 in the LVK population paper (arxiv2508.18083) is suggestive of a dip around $45 M_{\text{sun}}$. While this feature, obtained using the “B-Spline” model, is not measured with strong statistical significance, it may emerge as a clear feature in subsequent analyses.

4. Potential selection effects and χ_{eff} bias

The authors should consider and explain whether the apparent trend of χ_{eff} spanning a wider range at higher masses could arise from observational biases. For example, high-mass binaries have shorter inspiral durations in the LVK band, which naturally leads to broader posteriors on χ_{eff} estimates.

In our hierarchical Bayesian inference framework, we have carefully taken the selection effect and detector noise into account. In this framework, uninformative data doesn't push the posterior toward the transition spin model with a well measured spin transition mass. Rather, it simply yields bigger error bars without specific preference of either model. Thus, the fact that we are able to somewhat constrain the distribution of χ_{eff} for large masses cannot arise from the fact that more massive black holes tend to produce shorter signals.

Here's a simple proof illustrating our argument. Consider a completely uninformative likelihood: $L(d | \theta) = \text{constant}$. We analyse this uninformative data with a population model $\pi(\theta | \Lambda)$. Here, Λ are the population hyper-parameters. The population likelihood for N uninformative events is

$$\begin{aligned} L(d | \Lambda) &= \pi(\Lambda) \prod_i^N \int d\theta L(d_i | \theta) \pi(\theta | \Lambda) \\ &\sim \pi(\Lambda) \prod_i^N \int d\theta \pi(\theta | \Lambda) \\ &\sim \pi(\Lambda) \prod_i^N (1) \\ &\sim \pi(\Lambda) \end{aligned}$$

We see that the population likelihood returns the population prior when the data are uninformative.

5. Posteriors and high-spin claims

If the events within the mass gap originate from second-generation mergers, the individual BH spins are expected to be centered around $a=0.7$ (Gerosa & Berti 2017). The authors do not discuss the posteriors of the individual spins for these PISN events. I had a look and some cases shows peaks at lower values, around 0.4, making them unlikely products of second-generation mergers.

Given the limits of current detector sensitivity, the spin measurement of individual events is extremely uncertain—especially for parameters other than the effective spin. While some events have primary spin posteriors that peak below $\chi_1=0.7$, their large uncertainties make them consistent with a value of $\chi_1=0.7$. Thus the identification of black-hole formation channels through each individual event at current sensitivity is challenging. Instead, a hierarchical Bayesian inference is required to stack the information from all detections. This also allows us to take into account selection effects. (This is essentially the same point we make above in Q1 when discussing our ability to measure $\chi_{\text{eff}} < 0$ for individual events.)

In our newly added Section 1.9, we show that the χ_{eff} distribution is consistent with 2G+1G prediction: symmetric around 0 with half-width as 0.47. See also Fig. 2 of arxiv/2406.19044, which compares the high mass effective spin distribution with 2G+1G prediction, and Fig. 1 of arxiv/2303.02973, which suggests the spin magnitude for the high-mass mergers is consistent with 0.7.

6. Model testing strategy

The models explored in the manuscript appears tuned to reproduce the authors' preferred scenario (e.g., by effectively forcing a mass gap). To robustly assess the strength of the evidence, the authors should test a broader suite of alternative models. Crucially, the authors should quantify the relative support for each hypothesis and e.g., present the results in a summary table. Without such systematic model comparison, it is difficult to determine whether the data truly favor the proposed interpretation or whether the preferred model merely reflects prior/model choices.

In our method section we test several models that do not force a mass gap. In Section 1.3, we experiment with a notch filter model with a parameter A that controls the depth of the notch. While $A \sim 1$ effectively imposes an empty gap, the model allows for $A=0$ (no gap). In our Extended Fig. 3, we show the posterior of A is strongly peaked at 1, suggesting a strong preference for a gap. While the gap could be contaminated at some level (A slightly less than 1), we rule out $A < 0.09$ (at 99% credibility).

In Section 1.4, we present the results of a purely data-driven framework. We assume no mass model, and simply find the mass distribution that maximises population likelihood. The “ π stroke” plot in Extended Fig. 4 supports our result of a m_2 gap.

In Section 1.5, we further test the mass distribution using a pairing function formalism with a focus on the lower edge of the gap. Allowing the flexibility of the m_2 distribution to extend as high as 200 m_{sun} in the prior, the result shows the dearth of m_2 consistent with the mass gap model in the main text.

We thus conclude that our discovery of the mass gap is robust. As we reported in the main text, we calculate the Bayes factor of the model with and without the m_2 gap $\sim 10^2$. We

summarize the model comparison result using a few variations of our basic mass model in Table 1 including the Bayes factors and maximum likelihood values. For other tests where an apples-to-apples comparison using a simple Bayes factor calculation is not possible, we explain how they support our m2 gap discovery by text.

Minor Comments

- The upper-mass constraint (Extended Fig. 1) relies just on a single event.

We state this clearly in the main body of the manuscript on L120. We agree with the referee that one should be careful trying to interpret the upper edge of the gap. Thus, the focus of our work is the *lower* edge of the gap.

- Alternative formation pathways. The manuscript does not discuss well other formation channels, such as binary or triple interactions (possible spin-up, impact on secondary distribution). The authors do not explain well why the alternative scenarios are disfavored compared with the second-generation merger interpretation.

As far as we know, triple interactions do not provide a natural explanation for a contaminated gap. After receiving a strong recoil kick, it's very unlikely for the remnant of the merger of the first two components in a triple system to remain in the system and merge again with the third black hole. That said, we are happy for other authors to explore this possibility. The main finding of our paper is that there is a gap in the secondary mass distribution that coincides with a transition in the distribution of black hole spin. We argue that this can be naturally explained in terms of pair instability theory. We believe that hierarchical mergers offer a plausible explanation of the partial contamination of the mass gap, but we do not suggest that this is the only possible explanation for the contamination.

Referee #3 (Remarks to the Author):

I have one significant concern that the authors should address before I recommend the manuscript for publication in Nature. The fact that the mass model used `_requires_` there to be a gap in the secondary mass distribution is hidden in footnote 3 of the Methods section. While I understand that the Bayes factor gives a strong preference for the model with the gap, Bayes factors are hard to interpret since they are prior- and parameterization-dependent. The analysis should be repeated with a prior that allows for the width of the gap to go down to 0 Msun for a robust statement to be made on whether or not the data prefer the presence of a gap. An even better choice would be a slab-spike prior that gives equal prior support to the no-gap and gap hypotheses. I understand that this is partially addressed by the test with the notch filter of variable depth in Section 1.3, but the prior probability for an `_exactly_` continuous distribution is still 0 in this case. A plot of the posterior on the gap width, `w_g`, should also be included, as it's hard to tell how strongly a nonzero gap width is preferred from a plot of the joint posterior on the upper and lower gap edges. Such a prior choice that allows for a gap rather than enforcing its presence would make a much stronger argument for the gap model being preferred by the data. The Bayes factors alone are not convincing when the model enforces the presence of the gap.

We've rerun our mass gap model with the prior of the width to go down to 0 and updated Extended Figure 1 to include the result of the width of the gap. The data clearly rules out the width $< 40 M_{\text{sun}}$ at 99% credibility. We believe the referee's suggestion of a slab-spike prior is equivalent to the Bayes factor calculation already in the paper. In particular, our Bayes factor $\sim 10^2$ implies that the fraction of posterior support for the no gap model in a slab-spike prior would be $\leq 1\%$.

In our notch filter model, we allow for no gap when the depth parameter A is zero. That is, $A=0$ describes a continuous distribution without a gap. As we wrote above to Referee 2, "In our Extended Fig. 3, we show the posterior of A is strongly peaked at 1, suggesting a strong preference for a gap."

Finally, in Section 1.4 we show results obtained using the " π stroke" formalism. This analysis is entirely data-driven with no model for the mass distribution. The purple dots in Extended Figure 3 show the mass distribution, which maximises the population likelihood. The m_2 gap is clearly visible using this purely data-driven approach. This demonstrates that our parameterised models are not badly misspecified—the data prefer a gap-like feature.

Besides this main concern, I have a few detailed comments that I would ask the authors to consider.

- The discussion of GW190521 suggests that the most likely interpretation of this event given the inferred population model is as a straddling binary. The authors also find that their constraint on the upper edge of the mass gap hinges on GW231123. However, looking at Fig 3, it seems like there are two events with m_2 above the upper gap edge. Is this second event GW190521? Is GW190521 still a straddling binary when GW231123 is omitted from the population?

Yes, the second event is GW190521. Indeed, we believe that GW190521 can be interpreted as a straddling binary as in arxiv/2009.05472.

But as we clarified on L425, there is still marginal support for its secondary component from the far end of the gap. We've now added a bit details on L425

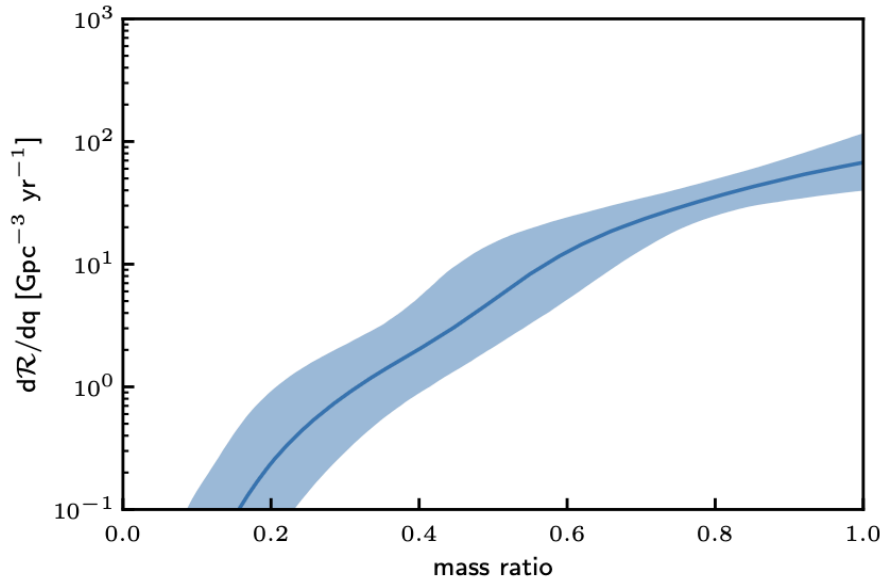
..., though there is still marginal support for m_2 at the other end of the gap $\sim 120 M_{\text{sun}}$. **There is posterior support $\sim 4\%$ for the hypothesis that both components of GW190521 are on the far side of the gap. Excluding GW231123, the overall population has a maximum $m_2 \sim 43 M_{\text{sun}}$ as shown in orange in our Extended Figure 1. Thus the event GW190521 in this case is fully consistent with its secondary component below the gap.**

- It should be clarified in the main text that the population model fits m_1/q rather than m_1/m_2 . This was not clear until the Methods section, where Eq 3 is presented. I also wonder what the resulting mass ratio distribution is, if there is such a strong gap in m_2 . Is it inconsistent with the typical power-law once marginalized over m_1/m_2 ? It would be interesting to see the PPD on mass ratio inferred under this population model.

We've edited the text to clarify the population model fits (m_1, q) in the main text on L108.

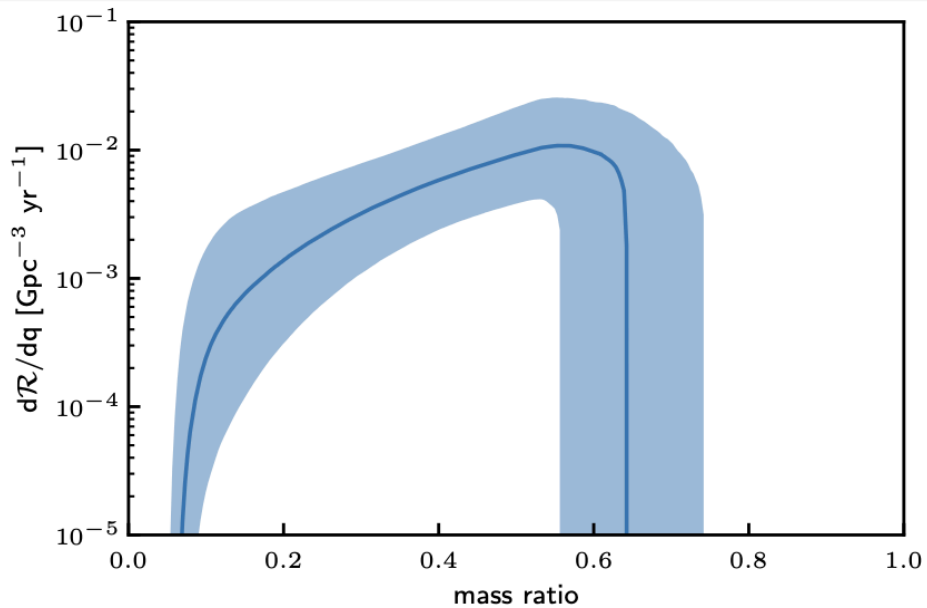
We employ a similar mass model formalism as in \cite{2508.18083} to describe the probability distribution $\pi(m_1, m_2)$ by fitting to the model of $\pi(m_1)$ and $\pi(q|m_1)$. However, we modify the model to allow for an interval corresponding to the pair-instability gap in m_2 , i.e., $m_1 \times q$, where the probability density is zero.

Here is the marginal distribution of mass ratio:



Although there's a significant gap in the secondary mass distribution, the marginal distribution for m_2/m_1 is smooth and resembles a power law.

We also made a plot showing a slice of the mass ratio distribution conditioned on $m_1 = 70M_{\text{sun}}$. The gap is clear in this case.



- Is there a typo in the x axis label of Extended Fig. 6, which just reads “The 99”? In any case, this quantity is missing units of M_{sun} .

Thanks for catching this. We have fixed the caption to read: “The 99% or 99.9% percentile of m_2 distribution [M_{dot}]”.

- The manuscript would benefit from a brief explanation of the astrophysical S-factor on which constraints are placed. Since many readers will not be nuclear physicists, it would be helpful to know what this parameter controls and how it affects the ^{12}C ^{16}O nuclear reaction rate.

We’ve now added a brief explanation on L213:

In astrophysics, the reaction cross section is commonly described by the S-factor, which is the nuclear structure-dependent part of the cross section for charged-particle nuclear reactions—isolated from Coulomb repulsion.

Referee #4 (Remarks to the Author):

Notes on Hierarchical Mergers:

The analysis in this paper depends very strongly on the assumption that 1G+2G BBHs are fairly common, while 2G+2G mergers are relatively rare. But most formation scenarios do predict *some* population of 2G+2G mergers, suggesting that, with more detections, the gap in m_2 will eventually be filled in as well. If one were to assume these mass-gap BHs originated from globular clusters, as is suggested here, then it is only a matter of luck that a 2G+2G merger has not been observed to pollute the gap. I would ask the authors to expand more on when (in terms of number of detections) we expect the m_2 gap to be similarly polluted, and whether this is a function of the specific formation channels (e.g. globular clusters, AGN disks, etc.).

To clarify, our analysis does not depend on the assumption that 2G+2G mergers are more rare than 1G+2G mergers. While our analysis allows for this possibility, it does not enforce it. For example, if the data preferred it, we would have recovered that there is no gap in m_2 . Instead, we find that the rate of BBH mergers with both components in the gap prefers to be zero, placing a very conservative upper limit of $< 0.0079 \text{ Gpc}^{-3} \text{ yr}^{-1}$ (while the rate of BBH mergers with one component in the gap is $\sim 0.25 \text{ Gpc}^{-3} \text{ yr}^{-1}$). Indeed, the referee is correct that we may eventually detect 2G+2G mergers with both components in the mass gap, allowing us to measure a nonzero rate that is still consistent with our upper limit.

Predictions from different formation channels, like globular clusters and AGN disks, provide context with which we can interpret these results. We briefly discussed the prediction of 2G+2G rate in globular clusters in our original manuscript, but the exact predicted rates depend on the black hole birth spins and the escape speed distribution of clusters, which are still uncertain. The prediction of hierarchical mergers in AGN is even more uncertain, but

their larger escape speeds may lead to a relatively higher rate of higher-generation mergers than we observe. We have expanded on our discussion on L145:

For example, the fraction of second-generation and first-generation black hole mergers (2G+1G) in **globular clusters** can be more than one order of magnitude higher than 2G+2G mergers [65]. Taking selection effects into account, 2G+2G mergers could constitute at most 1% of all detected binary black hole mergers if black holes are born with non-negligible spin [65], as hinted by other analyses [24, 51, 69]. The gap that we identify in the m_2 distribution may therefore represent the dearth of 2G+2G mergers. **The prediction of 2G+2G rates in AGN is much more uncertain, relying on the details of gas-assisted migration.**

Instead of discussing specific predictions from uncertain formation channels in more detail, we focus on the inferred rates of 2G+1G and 2G+2G mergers according to our model, which does not depend on assumptions specific to a formation channel. Our inferred rates are discussed in the paragraph on L214. Following the referee's suggestion, we have added a prediction of the number of 2G+2G detections in a typical catalog of 300 events based on our inferred mass distribution:

Assuming a GWTC-4 cumulative search sensitivity [85], we expect 53^{+25}_{-20} in 300 binary black hole events to have primary masses in the gap and secondary masses below the gap. Meanwhile, we infer 6^{+7}_{-5} in 300 binary black hole events to contain at least one component on the far side of the gap, out of which 0^{+3}_{-0} can be straddling binaries. **Under our best model, which indicates zero detections of binaries with both components in the mass gap, the Poisson counting uncertainty implies a 90% upper limit of $\lesssim 4.5$ such systems in a catalog of 300 binary black hole detections.**

Notes on PISN physics:

The authors provide a brief but good theoretical explanation of how the PISN process works and relating it to the features in the black hole mass spectrum. This is a satisfactory introduction to the theory of the problem.

The authors' model constrains the lower edge of the PISN mass gap for secondary (m_2) objects to $45 (+7/-6) M_\odot$. They remark that this is "remarkably consistent" with previous theoretical predictions. It should be noted that this prediction is highly reaction rate-dependent. Later figures in the paper reference the deBoer 2017 estimate for $12C$ α , but it should be noted that these theoretical papers use older rate estimates (Buchmann 1996; <https://ui.adsabs.harvard.edu/abs/1996ApJ...468L.127B/abstract>, Kunz et al., 2002; <https://ui.adsabs.harvard.edu/abs/2002ApJ...567..643K/abstract>) which disagree significantly. I contend that Figs. 4 and 8 should plot the S_{300} for these rates as well as the deBoer rate, if these theoretical predictions are going to be submitted for comparison.

We've added Buchmann 1996 and Kunz 2002 rate estimates in our Fig. 4 and Fig. 8 for reference now.

Furthermore, the authors' constraints for the S-factor of ^{12}C alpha diverges from the deBoer rate estimate significantly ($>+3\sigma$). Previous computational work on this problem using the median deBoer rate found a lower edge for the BH mass gap of $\sim 60M_{\odot}$ and did not compute models using reaction rates as high as the authors' estimate (Farag 2023). This fact itself warrants discussion, and it would be enlightening to see how the authors' estimates overlap with other sources for the ^{12}C alpha rate.

We've expanded our discussion at L230 as follows:

we constrain the astrophysical S -factor of $^{12}\text{C}(\alpha,\gamma)^{16}\text{O}$ at $\sim 300\text{ keV}$ to $\sim 256_{-104}^{+197}\text{ keV barns}$ (90% credibility), as seen in Fig.~\ref{fig:nuclear physics}. Our constraint on the S -factor is consistent with that of Ref.[86] with the median prediction of Ref.[86] within the 93% credible interval, though our posterior favours higher values. A higher S -factor is more consistent with the values reported in Refs.[87, 88, 90]. However, we do not interpret our result to be in tension with Ref.[86]. Since the mapping from the lower edge of the gap to the corresponding S-factor is non-linear (see Figure 8 of Ref.[12]), small shifts towards a lower inferred edge lead to a long tail towards higher inferred S -factors. We show in Sec.1.7 that shifting the lower edge by $\sim 2M_{\odot}$ results in a S -factor estimate that aligns well with the result of Ref.[86] (see the blue curve of Extended Fig. 8). Moreover, the simulations mapping from the gap to the corresponding $^{12}\text{C}(\alpha,\gamma)^{16}\text{O}$ rate in Ref.[21] simulated the lowest edge $\sim 40 M_{\odot}$ while our constraint has an error bar extending to $< 40M_{\odot}$. Thus, the long tail to high S -factors in our posterior may be unrealistic.

Minor/Textual comments:

Overall, the paper relies a little too heavily on citations to other analytic models. It would improve readability for a broader audience if the main features of these models were stated clearly (i.e. if some of the description from section 1.2 were moved into the main text).

We've added a bit more technical details of our mass gap model in our main text on L107-L110:

We employ a similar mass model formalism as in Ref.~\cite{GWTC_4_population} to describe the probability distribution $\pi(m_1, m_2)$ by fitting to the model of $\pi(m_1)$ and $\pi(q|m_1)$. However, we modify the model to allow for an interval corresponding to the pair-instability gap in m_2 , i.e., $m_1 \times q$, where the probability density is zero.

While section 1.2 also includes some other details about our overall mass models, they are not so relevant to our main focus of the m_2 gap.

We thank the referees for their thoughtful comments. We reply below with the referee comments in black and our own replies in purple. We highlighted our changes in our latest manuscript in blue.

Referee #1 (Remarks to the Author):

The authors have carried out a solid and careful analysis, exploring the robustness of a putative mass gap in the secondary-mass distribution of binary black holes. In my view, they have done essentially what is possible with the current gravitational-wave catalog. The manuscript presents a coherent and plausible scenario that, given the present dataset, offers a promising interpretation of the observations. The additional tests included in the revised version provide meaningful cross-checks and further support the conclusion that a genuine trough or gap in m_2 is preferred by current detections.

That said, several aspects of the interpretation remain limited more by the quality and quantity of the available data than by any shortcoming in the authors' analysis. For example, the evidence for a fully isotropic spin distribution in the high-mass population is consistent with isotropy but does not demonstrably favour it over mildly positive-shifted alternatives; the individual spin posteriors for putative second-generation black holes are too broad to support the characteristic $a \sim 0.7$; and the absence of a feature at $m_1 \approx 45 M_\odot$ can be accommodated within their scenario, but such an explanation is intrinsically model-dependent.

While the pair-instability interpretation proposed by the authors is certainly a promising explanation, I am not fully convinced that, given the still small-number statistics, alternative scenarios can be ruled out. In particular, a standard initial mass function (IMF, which limits the number of massive stars), either alone or combined with non-conservative binary interactions, could produce the reported cut-off in the secondary-mass distribution, while high-mass χ_{eff} features could also be shaped by selection effects or other astrophysical mechanisms. Similar caveats apply to the flexibility in assigning several candidate events to the putative PSN mass gap.

Moreover, I note that a recent preprint (Ray & Kalogera 2025, arXiv:2510.18867) appears to challenge some of the claims regarding a sharp mass gap in the secondary-mass distribution. The authors could strengthen their manuscript by explicitly discussing this work, clarifying potential points of agreement or tension, and explaining whether their conclusions remain robust in light of these results.

I therefore recommend that the manuscript be revised to:

1. Moderate the strength of claims regarding the definitiveness of the PSN interpretation or try to quantify alternative explanations (e.g., the role of IMF in m_2 limit);

Following the referee's suggestion, we've added the following discussion at the end of the main body

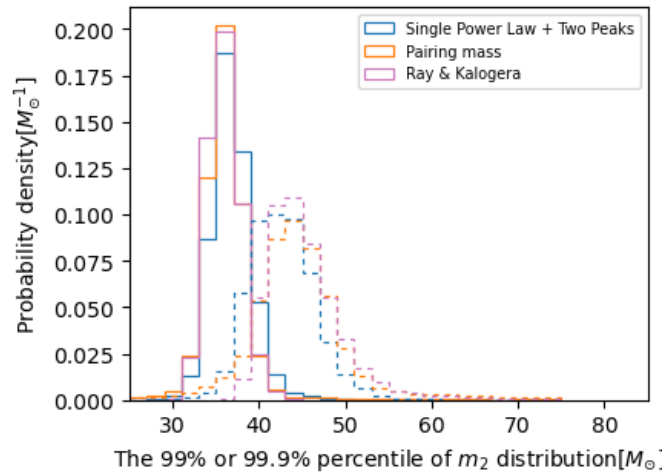
While we interpret the dearth of binary black holes with $m_2 \gtrsim 45 M_\odot$ as evidence of a pair instability gap---and while we regard this explanation as a promising and plausible explanation---we acknowledge that the data are open to other interpretations. For example, one could imagine that a dearth of events with

$m_2 \gtrsim 45 M_\odot$ might be produced through the complicated interplay between the initial stellar mass function and binary interactions, thereby mimicking a pair instability gap through unrelated physical processes. If subsequent data continue to show a gap for binaries with $m_2 \gtrsim 45 M_\odot$ regardless of the mass of the primary black hole, alternative explanations of the gap invoking binary interactions may begin to appear contrived. More excitingly, future measurements of spin orientations may help indicate the evolutionary pathways and formation environments of high-spin events to further test the hierarchical merger hypothesis.

We comment on the preprint by Ray & Kalogera in our reply to point 2.

2. Include a discussion of the recent preprint by Ray & Kalogera (2025, arXiv:2510.18867), addressing any potential tension or reconciliation with their results.

Our position is that the results in Ray & Kalogera are not actually in tension with ours. Their analysis on mass ratios shows a clear transition in the mass ratio distribution around 40 $M_\odot$ (green curve in their Figure 2). Their power law index of the mass ratio distribution changes from positive (merger rate increases as mass ratio increases) to negative (merger rate increases as mass ratio decreases) after this mass scale. This indicates the dearth of high m_2 mergers, consistent with our discovery. We have reanalysed the data using the model from Ray & Kalogera. We excluded GW231123, focusing on the lower edge of the gap, and compared the result with that from our Extended Figure 6. Their result matches ours pretty well. In the below figure, the solid lines show the 99th percentile of the m_2 distribution, and the dashed lines show the 99.9th percentile.



We were asked by the Editor of the Astrophysical Journal Letters to provide comments on the manuscript by Ray & Kalogera. We attached our comments below for reference.

We were contacted by the Editor of the Astrophysical Journal Letters to provide a comment on this manuscript, “Reexamining Evidence of a Pair-Instability Mass Gap in the Binary Black Hole Population.” The context of this request is that the manuscript addresses claims made in our recent paper, “Evidence of the pair instability gap in the distribution of black hole masses.”

While we have a somewhat different perspective than Ray & Kalogera, we have no objection to some version of the paper being published once the authors have addressed comments from the referees, whatever those may be.

The main contention of our paper is that there's a gap in the distribution of the secondary (lighter) black holes observed by LIGO-Virgo-KAGRA and that this gap is coincident with a qualitative shift in the distribution of black hole spin. The manuscript by Ray & Kalogera challenges this conclusion by checking for *model misspecification*. That is, they want to determine if our conclusions are driven by the data or by the assumptions in our model. In general, this is a useful thing to do—especially following a bold claim—and so we welcome the scrutiny.

Using a different model, Ray & Kalogera find that the gap may not be as pronounced as we find it to be in our primary model. We employed several flexible models in our study to validate the robustness of our conclusion. We first experiment with a notch filter model with a parameter A that controls the depth of the notch. While $A \sim 1$ effectively imposes an empty gap, the model allows for $A=0$ (no gap). In our Extended Fig. 3, we show the posterior of A is strongly peaked at 1, suggesting a strong preference for a gap. We further employ a pairing function formalism which fits the secondary mass distribution to a Broken Power Law + Two Peaks distribution. This model is similar to the model in Ray & Kalogera which allows a break point in secondary mass distribution. Our result using this more flexible model shows a dearth of high secondary mass mergers, consistent with the conclusion of our primary model (see our Extended Fig. 6), although the sharpness of the gap can not be constrained in this highly flexible model. From our perspective, the details of the gap (e.g., the extent to which it is completely empty) are less important than the fact that there is *some* kind of gap. We contend that both papers show evidence of gap-like structure beginning at 45 solar masses. Thus, from our perspective, the conflict presented by Ray & Kalogera might be somewhat exaggerated. However, this may be in the eye of the beholder.

Moreover, the coincidence between the spin transition and the m_1 transition gives us extra confidence that something special happens at 45 solar masses as also shown in Figure 4 of Ray & Kalogera. Antonini et al. (ariv/2406.19044) has thoroughly investigated the spin properties of those high-mass mergers and found them consistent with 2G+1G prediction.

Finally, our result is supported by a purely data-driven framework: the “ π stroke” formalism, which explores the maximum population likelihood fit without any prior assumptions. The fact that we see a gap in our π stroke reconstruction (Extended Fig. 4) proves definitively that the data prefer a gap. This shows unequivocally that the gap we find does not arise entirely from model misspecification.

In sum, this work by Ray & Kalogera does not change our interpretation of the results from Tong et al. However, we think that their investigation into misspecification is worthwhile. We are happy to agree to disagree about the extent to which the two papers are actually in conflict. We have included some additional suggestions in the postscript below.

Best regards,
Hui Tong, Maya Fishbach, Eric Thrane

P.S. Here are some additional comments for the author's consideration.

- While there is a break point in the secondary mass distribution clearly constrained in Ray & Kalogera, the authors argue that the inferred mass distribution is not consistent with the interpretation of a gap but rather a smooth fall-off. The main quantitative support for this statement is the $m_2, 99\%$ (or $m_2, 99.9\%$) posterior in Figure 2. The definition of this quantity is the 99% percentile of m_2 distribution within $m_{2,b1} < m_2 < m_{2,b2}$. In our study, we have shown that there is marginal support for high m_2 mergers, possibly with $\leq 1\%$ mergers between $[m_{2,b1}, m_{2,b2}]$. Thus the 99% (99.9%) percentile within this interval might represent $99\% + 1\% \times 99\%$ (or 99.9%) = 99.99% (or 99.999%) percentile of the overall BBH population excluding GW231123. It's unclear how well one can constrain this quantity given just ~ 150 detections. The manuscript would benefit from discussion about how much their constraints on this quantity is driven by data or by their model/prior choice. One naive calculation is to show the prior of this quantity given their inferred $m_2, b1$ and $m_2, b2$.
- The authors argue that the properties of the high-mass subpopulation are inconsistent with the hierarchical merger prediction mainly by the inferred mass ratio distribution. The right panel of Figure 4 seems to support the interpretation of inconsistency. However, the motivation of the choice of the mass ratio threshold ~ 0.6 for the comparison with CMC result is unclear. By looking at the cumulative density (the center panel of Figure 4), the observational constraint in green is much more uncertain than the theoretical prediction in orange. The posterior in green is so broad that the comparison with the prediction in orange is quite sensitive to the choice of threshold. While the choice of mergers with mass ratio > 0.6 would imply the observational population has more equal-mass systems, the comparison based on a different choice of threshold for mergers with mass ratio < 0.3 would actually lead to the opposite conclusion. One could argue that the posterior in green of the observations indicates more unequal mass systems with mass ratio < 0.3 compared with CMC simulations. By eye, a typical threshold of mass ratio ~ 0.5 for 2G+1G mergers would result in a conclusion in-between. The number of events with mass ratio > 0.5 seems to be consistent with hierarchical merger prediction. The manuscript would benefit from a discussion of the motivation of their choice of the mass ratio threshold ~ 0.6 . As the authors have noticed, Banagiri et al. has looked into the mass ratio distribution of different subpopulations and found the mass ratio distribution for those high mass mergers is consistent with ~ 0.5 . While the result on spin properties from Banarigi et al. might not support the hierarchical interpretation, their mass ratio result supports the 2G+1G interpretation of a subpopulation with mass ratio ~ 0.5 . Thus the statement in Ray & Kalogera about Banarigi et al. on the first paragraph of page 7 of the arxiv version is at least incomplete:

Our conclusions are in principle, consistent with the findings of S. Banagiri et al. (2025), who characterize the distributions of component spin magnitudes and tilts in this high-mass sub-population, and find features (such as support for low primary spins, high secondary spins and no transition in the spin tilt distributions) that are also difficult to explain exclusively with the (2G+1G) hierarchical merger scenario.

We'd also like to note that, while Ray & Kalogera raise questions about the existence of a gap in the secondary mass spectrum, we are aware of at least two recent papers which *support* our conclusion. Here are the titles and authors:

“Signatures of a subpopulation of hierarchical mergers in effective precessing spin” by Cailin Plunkett, Salvatore Vitale, Thomas Callister, Michael Zevin, <https://arxiv.org/abs/2601.07908>

Farah, Vijaykumar & Fishbach (authored by some of us) <https://arxiv.org/abs/2601.03456>; especially their Figure 2.

“Gravitational waves reveal the pair-instability mass gap and constrain nuclear burning in massive stars” by Antonini et al., <https://arxiv.org/pdf/2509.04637>, who claimed the PISN evidence by spin features.

Referee #2 (Remarks to the Author):

I thank the authors for addressing my previous comments and concerns. I find the new posterior on the gap width shown in Extended Fig. 1 to be particularly compelling evidence for the presence of the mass gap in the m_2 distribution. The current manuscript is clear, provides strong justification and validation of the mass gap conclusion, and presents one of the most significant results in gravitational-wave population inference to date, revealing a feature that has long been predicted in astrophysical models. I recommend the manuscript for publication in Nature. Below, I include some stylistic comments for the authors' consideration.

- The revised introduction of the parameterization used for the population model is confusing. The main population model considered in this work does not directly model the distribution of $\pi(m_1, m_2)$, so keeping this in L108 seems misleading to me. Waiting to define the mass ratio q until the sentence after it is introduced also comes across as disjointed. I encourage the authors to revise these sentences to improve the clarity and flow.

We thank the referee for their suggestion. We've revised the description of the model parameterization in the main body to help avoid any confusion.

... describe the probability distribution $\pi(m_1, m_2)$ by fitting to the model of the primary mass distribution $\pi(m_1)$ and the conditional secondary-mass distribution $\pi(m_2|m_1)$.

The difference between $\pi(m_2|m_1)$ and $\pi(q|m_1)$ is only a Jacobian term, $\pi(m_2|m_1) = \pi(q|m_1) / m_1$. The distribution of the secondary mass distribution is also a power-law distribution. Thus we've also changed the description at the appendix accordingly:

As discussed below, we employ various models for $\pi(m_1|\Lambda)$ from Ref.~\cite{GWTC_4_population}. The key ingredient, however, is our model for [secondary mass](#), which enforces a gap in the distribution of m_2

$$\pi(m_2|m_1, \Lambda) \propto \begin{cases} 0 & m_g \leq m_2 \leq m_g + w_g, \\ m_2^{\beta_q} & \text{otherwise.} \end{cases} \quad (3)$$

- I would encourage the authors to add the fact that the absence of a gap is ruled out at high significance (by eyeballing the new Extended Figure 1, it seems like $> 5\sigma$) to the paragraph at the bottom of page 3. This, to me, is much more convincing than a lnBF preference of ~ 4.5 .

We've added the corresponding statement at the bottom of page 3:

The absence of a gap, i.e., the width of the gap to be $< 20 M_\odot$, is ruled out at 99.9% credibility(see Extended Figure~\ref{fig:corner_m2gap}).

- Some typos in the new L187: While hierarchical mergers in dense stellar clusters are usually predicted to have isotropic spin **orientations** due to dynamical assembly, preferentially aligned spins can arise from black holes **in AGN disks** (?) or interactions between black holes and stars in clusters [49, 83].

Fixed.

- L202 quotes $\ln\text{BF} > 8$ as the threshold for “strong support,” but the Bayes factors between population models in the text and in the new Table 1 are not given as lnBFs. Since the log is easier to interpret in terms of significance, I recommend that the authors switch to using lnBFs to compare the preference between population models.

We've switched to using lnBFs throughout the manuscript.

- Some typos in L300: "The slopes of the two bands of **the** broken power law distribution are consistent with each other. We interpret the initial requirement of a second power law in **the** m_1 distribution in current data as a projection of the dearth of high m_2 events. **Instead,** ... “

Fixed.

- Please add lnBFs comparing the evidence for the pairing mass model and spin transition models (Eq 7/8) against the other models to Table I.

For the pairing mass analysis, we focus on the investigation of the lower edge by excluding GW231123. (Bayes factors are only defined for a fixed dataset, so we cannot compare the evidence for one model including GW231123 to another model in Table 1 excluding GW231123.) For the spin transition models (Eq 7/8), we construct the model in terms of the effective inspiral spin (χ_{eff}), but Table 1 shows models where the spin distribution is parameterized in terms of the component spins (a_1 , a_2 , $\cos_{\text{tilt1}}$ and $\cos_{\text{tilt2}}$), as we introduced in L294. There are four spin parameters in the models described by component spins but only one parameter, χ_{eff} , in the models described by the effective inspiral spin. The different degrees of freedom in these models make direct comparison of Bayesian evidence complicated. Rather than comparing these models directly using Bayes factors, we instead justify the spin transition model as follows. In footnote 2, we explain that if we model the spin distribution in terms of effective spins, a model with a spin transition is strongly favored over a model without a spin transition (in which the effective spin distribution is independent of mass, see also Pierra et al. <https://arxiv.org/abs/2406.01679> which supports the preference of spin transition by component spin parameterization). And separately, we present the Bayes factor between a model with a spin transition and an m_2 gap compared to a model with a spin transition but without an m_2 gap in L181. We believe these comparisons sufficiently show that a model with both a spin transition and an m_2 gap is preferred.

- I suggest moving the new Section 1.9 to immediately follow Section 1.6, since they are both focused on the spin transition model and the spin of the high-mass subpopulation.

We've rearranged the order following this suggestion.